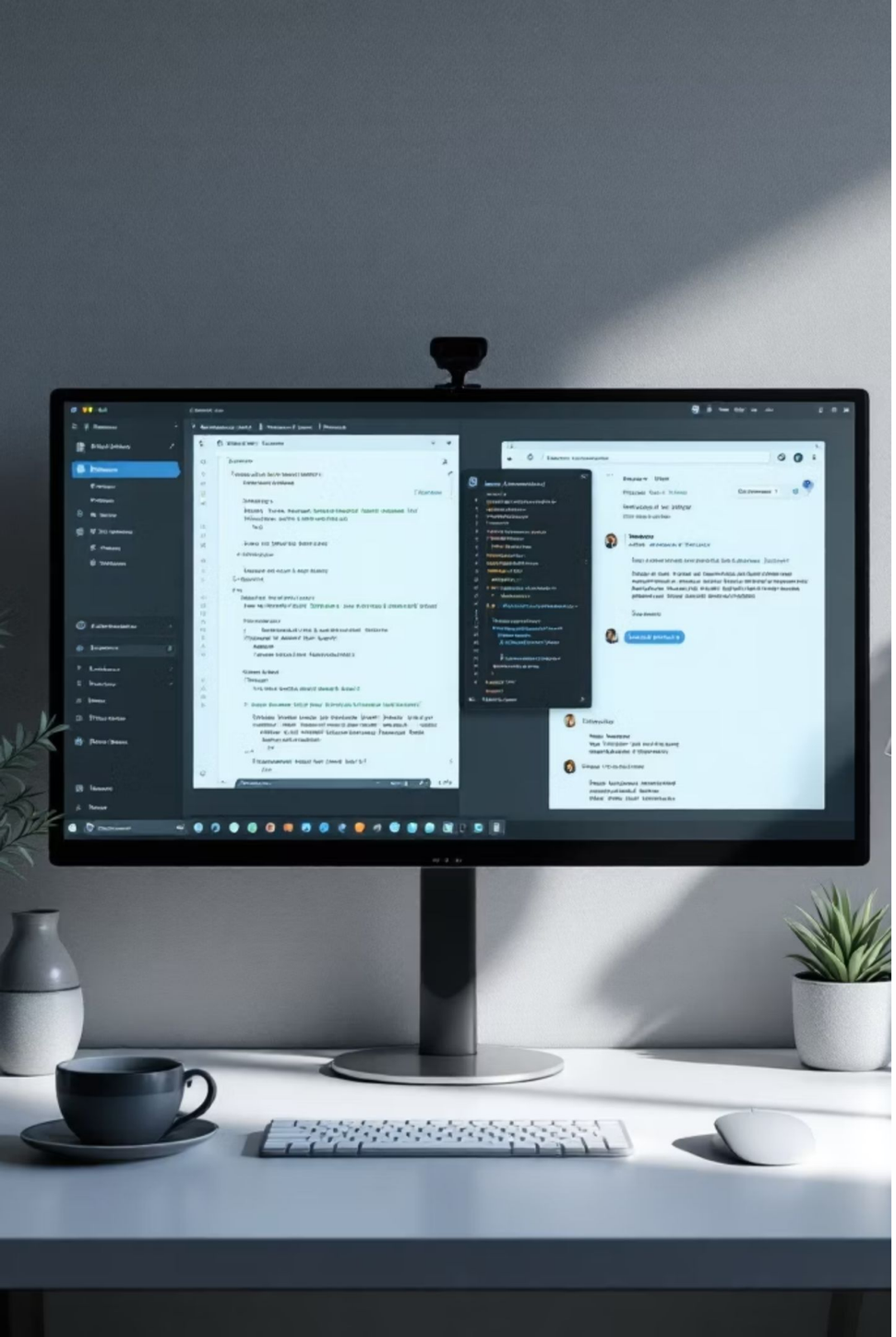
SECAI+ LESSON 5.3.3

Automate Security Tasks

Discover how to harness AI tools like ChatGPT to generate, customize, and convert security scripts across programming languages, transforming natural language prompts into fully functional security automation tools.



Using ChatGPT for Script Generation

ChatGPT (OpenAI) enables security professionals to generate complete Python security scripts directly from natural language prompts with no manual coding required from scratch.

What AI Provides

- Overview of the script's purpose
- Prerequisites & dependencies
- Full source code
- Usage instructions

Platform Adaptability

- Windows environments
- Linux systems
- macOS environments
- Cross-platform customization

Existing Script Analysis

- Feed existing scripts into AI
- Receive improvement suggestions
- Identify vulnerabilities in code
- Optimize performance

Demo 1 — OSINT Footprinting Script

Goal: Gather open source intelligence about a public domain (e.g., `comptia.org`) using an AI-generated Python script called `footprinter.py`.

DNS Lookups

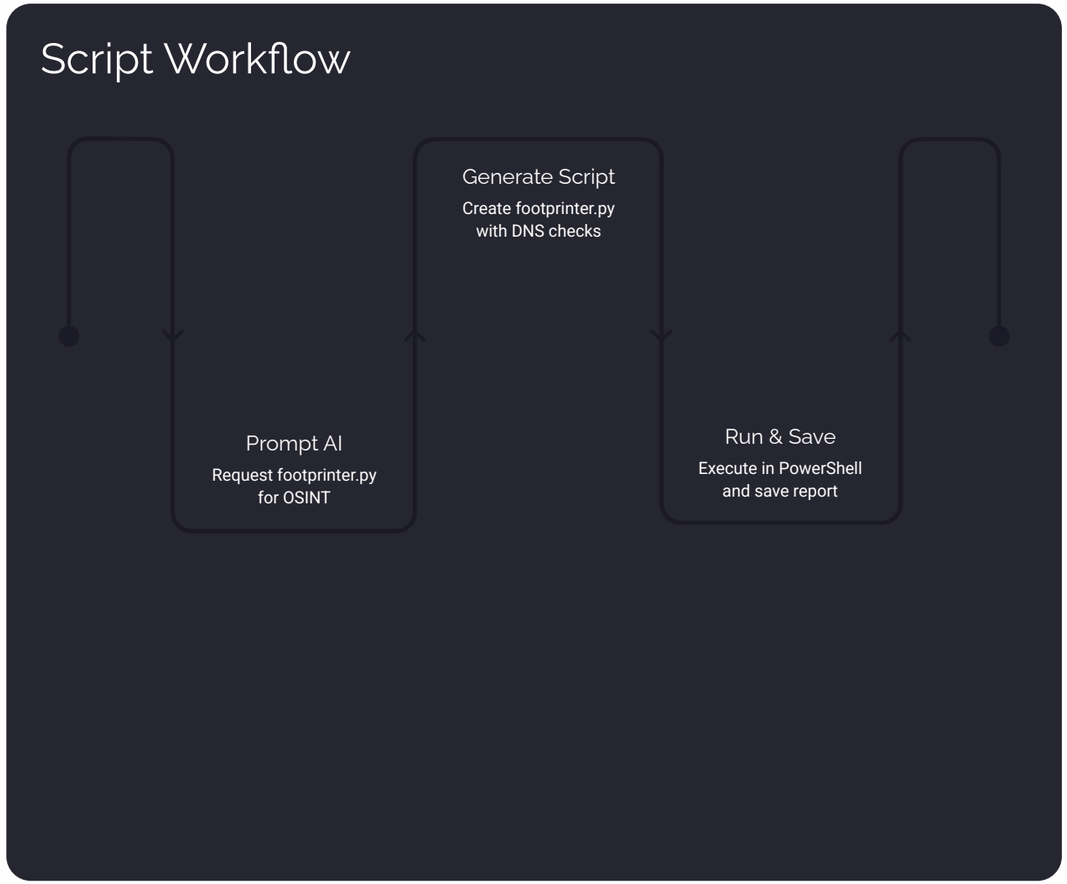
Forward and reverse DNS resolution to map domain infrastructure.

Email Enumeration

Harvests publicly available email addresses from the web.

Execution & Output

Executed via PowerShell; saves the full report to the local desktop.



Demo 2 — Network Scanning Script

Using AI-generated scripts powered by **NMAP**, security teams can discover live systems, open ports, running services, and service versions across an entire network, all exported cleanly to CSV.

Full Port Coverage

Scans all **65,535 ports** to ensure no open service goes undetected.

Structured CSV Output

Results exported with columns: IP address, hostname, protocol, port, service, and version.

On-the-Fly Customization

Modify the AI prompt to instantly change the IP range or scan parameters without rewriting code.

Converting Scripts Across Languages

One of the most powerful capabilities of AI-assisted scripting is **instant language conversion** — the same security logic can be translated between programming languages in seconds, making tools immediately deployable across any environment.

1

Python

Original script — cross-platform baseline

2

PowerShell

Converted for Windows environments

3

Bash

Converted for Linux & macOS environments

AI-powered language conversion **dramatically reduces** the time needed to adapt security tools for different operating environments — no manual rewriting required.

Key Takeaways

AI-powered security scripting represents a fundamental shift in how security professionals build and deploy automation tools.

1. **Complete Scripts from Natural Language**
   ChatGPT and AI tools generate fully functional security scripts from plain English prompts — no deep coding expertise required.

2. **Step-by-Step Guidance Included**
   Every AI-generated script comes with setup instructions, prerequisites, and usage guidance built in.

3. **Fully Customizable via Prompt**
   Scripts are instantly adaptable — simply modify the prompt with specific conditions, IP ranges, or parameters to change behavior.

4. **Instant Cross-Platform Conversion**
   Python, PowerShell, and Bash conversions happen in seconds, making any tool immediately cross-platform without manual rewriting.

5. **Analyze & Improve Existing Scripts**
   AI can review and enhance internally developed scripts, identifying improvements and optimizations proactively.